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Agentic AI for Disaster Management System

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Guide Signature with date

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ABSTRACT:

Urban regions are increasingly exposed to natural and human-induced disasters due to rapid urbanization, climate change, and complex infrastructure systems. Conventional disaster response mechanisms are largely reactive, centralized, and dependent on human decision-making, which often results in delayed response times, inefficient coordination, and suboptimal utilization of emergency resources. Recent advancements in artificial intelligence have enabled the development of autonomous and intelligent systems capable of operating in dynamic and uncertain environments. Among these, Agentic AI introduces a paradigm where multiple autonomous agents collaboratively perceive, reason, decide, and act to achieve predefined goals. This project proposes an Agentic AI-based framework for Disaster Response and Urban Emergency Management to enhance preparedness, response efficiency, and situational awareness.

The proposed system, named **AGENT-RESCUE**, integrates multiple AI agents responsible for disaster prediction, situation assessment, resource allocation, communication, citizen assistance, and continuous learning. Machine learning models are employed to predict disaster risks, while natural language processing and computer vision techniques are used to assess real-time emergency information. Reinforcement learning is applied to optimize emergency resource deployment. The system is implemented using a Python-based full-stack architecture with FastAPI and real-time communication mechanisms. Explainable AI techniques are incorporated to ensure transparency and accountability in decision-making. The proposed solution demonstrates improved response time, coordination, and adaptability, making it suitable for modern smart city environments.

Keywords: Agentic AI, Disaster Management, Emergency Response, Multi-Agent Systems, Smart Cities

INTRODUCTION:

Disaster response and urban emergency management play a vital role in safeguarding human life, infrastructure, and economic stability. Urban disasters such as floods, fires, earthquakes, industrial accidents, and medical emergencies require rapid and coordinated responses from multiple agencies. However, existing disaster management systems rely heavily on manual coordination and static rule-based decision-making, which often fail under high-pressure and rapidly evolving situations. The increasing frequency and intensity of disasters highlight the need for intelligent systems capable of handling complex emergency scenarios efficiently.

Artificial intelligence has emerged as a powerful tool for improving disaster preparedness and response. Recent research has shown that machine learning, deep learning, and multi-agent systems can significantly enhance prediction accuracy and decision-making speed. Agentic AI extends these capabilities by enabling autonomous agents to operate with specific goals, environmental awareness, and learning mechanisms. By applying Agentic AI to urban emergency management, this project aims to develop a system that can autonomously predict disasters, assess severity, allocate resources optimally, communicate effectively with stakeholders, and continuously improve through feedback. This approach represents a shift from reactive systems to proactive and adaptive disaster management solutions.

PROBLEM STATEMENT:

Urban disaster management systems currently face significant challenges in handling emergencies efficiently and effectively. Most existing systems are reactive in nature, triggering alerts only after a disaster has occurred. They lack real-time situational awareness and depend heavily on human operators for critical decision-making. During large-scale emergencies, this dependency leads to information overload, delayed responses, and poor coordination among emergency services such as hospitals, fire departments, police units, and disaster relief organizations.

Furthermore, traditional systems do not incorporate learning mechanisms to improve future responses based on past experiences. Decisions made during emergencies are rarely analyzed systematically, resulting in repeated inefficiencies. The absence of intelligent resource allocation leads to uneven distribution of emergency services, causing delays in critical areas while resources remain underutilized elsewhere. There is also a lack of transparency in decision-making, which reduces trust in automated systems. Hence, there is a strong need for an intelligent, autonomous, explainable, and adaptive disaster management system that can predict potential disasters, assess real-time emergency conditions, allocate resources optimally, coordinate communication, and learn continuously to enhance urban resilience.

OBJECTIVES:

The primary objective of this project is to design and implement an **Agentic AI–based Disaster Response and Urban Emergency Management System** that can autonomously predict, analyze, decide, and respond to emergency situations in urban environments. To achieve this primary goal, the following specific objectives are defined:

To develop a multi-agent Agentic AI framework for disaster management

This objective focuses on designing a system composed of multiple autonomous AI agents, where each agent is assigned a specific responsibility such as prediction, assessment, allocation, communication, and learning. The agents operate independently but collaborate through shared information to handle complex and dynamic emergency scenarios efficiently.

To predict potential disasters using machine learning techniques

The system aims to analyze historical disaster records and real-time environmental data to predict the probability, type, and severity of disasters. This enables early warnings and proactive preparedness rather than reactive responses.

To assess real-time emergency situations using intelligent analysis techniques

Natural Language Processing (NLP) is used to analyze emergency messages, reports, and citizen inputs, while Computer Vision techniques analyze visual data such as images or CCTV feeds. This helps in accurately identifying the urgency and impact of the emergency.

To optimize emergency resource allocation using reinforcement learning

The project aims to minimize response time and maximize efficiency by intelligently allocating emergency resources such as ambulances, fire trucks, police units, hospitals, and shelters. Reinforcement learning enables the system to learn optimal deployment strategies based on past outcomes.

To provide real-time communication and decision support

The system facilitates automated alerts, instructions, and coordination among emergency authorities, healthcare facilities, relief organizations, and citizens. This reduces misinformation and improves coordination during critical situations.

To incorporate explainable and trustworthy AI mechanisms

The system ensures transparency in AI decisions by providing explanations for predictions and resource allocation decisions. This builds trust, accountability, and ethical compliance in public safety applications.

To enable continuous learning and system improvement

Post-disaster analysis is used to evaluate system performance and retrain models, ensuring continuous improvement and adaptability to new disaster patterns.

SCOPE OF THE PROJECT:

The scope of this project encompasses the design, development, and evaluation of an intelligent disaster response system tailored for **urban emergency management**. The system is designed to handle various types of disasters including floods, fires, earthquakes, industrial accidents, road accidents, and medical emergencies. It supports real-time monitoring, intelligent decision-making, and coordinated response across multiple emergency services.

The project scope includes the implementation of a **full-stack software architecture**, consisting of a web-based dashboard for administrators and emergency authorities, and an interactive interface for citizens. The system integrates multiple AI techniques such as machine learning for prediction, natural language processing for textual data analysis, computer vision for visual assessment, and reinforcement learning for optimal resource allocation. The use of Agentic AI allows autonomous operation while maintaining human-in-the-loop control for critical decisions.

The system is designed to be **scalable and extensible**, allowing future integration with smart city infrastructure such as IoT sensors, weather stations, satellite data, drones, and mobile applications. Although the current implementation uses simulated and publicly available datasets, the architecture supports real-time data streams and cloud deployment. Ethical considerations, data security, explainability, and system reliability are also within the scope of this project. However, physical deployment of hardware sensors and real-time governmental integration are considered outside the scope and are proposed as future enhancements.

PROPOSED SYSTEM:

The proposed system, **AGENT-RESCUE**, is an **Agentic AI–based Disaster Response and Urban Emergency Management System** designed to operate autonomously while supporting human decision-makers during emergency situations. The system follows a structured end-to-end workflow starting from disaster detection to post-event learning and improvement. It is composed of multiple intelligent agents, each performing a specialized task and collaborating through a centralized backend infrastructure.

1. Data Collection and Continuous Monitoring

The system continuously collects data from multiple sources such as historical disaster databases, real-time weather information, sensor feeds (simulated), emergency reports, and citizen-generated data. This data includes structured data (numerical and categorical) as well as unstructured data (text messages and images). Continuous monitoring ensures that early warning signs are detected before a disaster escalates.

2. Disaster Prediction and Risk Analysis

The Disaster Prediction Agent analyzes the collected data using machine learning models to predict the likelihood, type, and severity of potential disasters. The agent generates risk scores and identifies high-risk zones within urban areas. If the predicted risk exceeds a predefined threshold, the system automatically triggers alerts and initiates further assessment processes.

3. Situation Assessment and Severity Evaluation

Once an alert is generated, the Situation Assessment Agent evaluates the real-time emergency conditions. Natural Language Processing techniques are used to analyze emergency messages, reports, and social media inputs to detect urgency and intent. Computer Vision models analyze images or video frames to identify visual indicators such as fire, smoke, flooding, or structural damage. The agent assigns a severity level that determines the urgency of response.

4. Intelligent Resource Allocation and Coordination

The Resource Allocation Agent is responsible for optimizing the deployment of emergency resources such as ambulances, fire trucks, police units, hospitals, and shelters. Reinforcement learning techniques are used to select optimal resource deployment strategies based on factors like distance, availability, severity, and response time. The agent continuously learns from previous decisions to improve future performance.

5. Communication, Assistance, and Continuous Learning

The Communication Agent disseminates real-time alerts, emergency instructions, and coordination messages to authorities, hospitals, NGOs, and citizens. A chatbot interface provides guidance, first-aid instructions, and shelter information to the public. After the emergency, the Learning and Feedback Agent analyzes system performance, evaluates decision outcomes, and retrains models to enhance future responses. This ensures continuous improvement and adaptability.

LITERATURE SURVEY:

S.NO	TITLE	MERITS	DEMERITS
1	AI-Based Disaster Response Systems (2021)	Early disaster prediction improves preparedness; Automated alerts reduce response delay	Limited adaptability to new disaster patterns; Lack of autonomous decision-making
2	Multi-Agent Systems for Emergency Management (2022)	Decentralized control improves coordination; Parallel task execution	High computational complexity; Difficult agent communication
3	Deep Learning for Flood Prediction (2021)	High prediction accuracy; Handles complex patterns	Requires large datasets; Poor generalization

4	Social Media Analytics for Crisis Detection (2020)	Real-time situational awareness; Public sentiment analysis	High noise in data; Risk of misinformation
5	Reinforcement Learning for Resource Allocation (2023)	Optimized resource usage; Learns from experience	Slow training process; Requires careful reward design
6	Computer Vision-Based Fire Detection (2022)	Fast visual detection; Useful for early alerts	False positives; Sensitive to lighting conditions
7	Explainable AI in Public Safety Systems (2020)	Improves trust; Enhances accountability	Reduced model performance; Added complexity
8	Smart City Emergency Management Platforms (2021)	Scalable infrastructure; Integrated services	High deployment cost; Infrastructure dependency
9	IoT-Enabled Disaster Monitoring Systems (2022)	Real-time sensing; High data accuracy	Security vulnerabilities; Maintenance overhead
10	NLP-Based Emergency Chatbots (2021)	Improves citizen assistance; Reduces panic	Language limitations; Context misunderstanding
11	Autonomous Decision Support Systems (2021)	Reduces human workload; Faster decisions	Ethical concerns; Limited transparency
12	Hybrid AI Models for Disaster Management (2022)	Improved performance; Combines multiple techniques	Complex system design; Difficult maintenance
13	Cloud-Based Emergency Response Systems (2021)	High availability; Scalable resources	Latency issues; Data privacy risks
14	Ethical AI	Ensures responsible AI use;	Difficult implementation;

	Frameworks for Disaster Management (2020)	Improves trust	Lack of standardization
15	Urban Resilience and AI Frameworks (2023)	Long-term disaster preparedness; Policy support	Slow adaptability; Limited automation

FINDINGS IN LITERATURE SURVEY:

Strengths Identified in Existing Literature:

Effective Use of AI for Disaster Prediction

Several studies demonstrate that machine learning and deep learning models significantly improve the accuracy of disaster prediction, particularly for floods, earthquakes, and fires. These models can process large volumes of historical and real-time data to identify hidden patterns.

Improved Coordination through Multi-Agent Systems

Research on multi-agent systems highlights their ability to decentralize decision-making and handle multiple emergency tasks in parallel, improving coordination among emergency services.

Real-Time Situational Awareness

The use of social media analytics, IoT sensors, and computer vision techniques provides real-time insights into emergency situations, enabling quicker detection and response.

Optimization through Reinforcement Learning

Studies show that reinforcement learning improves emergency resource allocation by learning optimal strategies based on past outcomes rather than relying on fixed rules.

Growing Importance of Explainable AI

Literature emphasizes the need for transparency and accountability in AI-based public safety systems, especially where decisions directly impact human lives.

Limitations Identified in Existing Literature:

Lack of Autonomous End-to-End Decision-Making

Most existing systems focus on prediction or monitoring but do not autonomously take decisions and execute actions across the full disaster response lifecycle.

Limited Learning and Adaptability

Many systems do not incorporate feedback mechanisms to learn from past disasters, resulting in repeated inefficiencies and static behavior.

High Dependence on Human Operators

Decision-making in critical stages such as resource allocation and coordination often requires manual intervention, increasing response time during emergencies.

Poor Explainability in Decision Processes

While some systems use AI models, they do not clearly explain why specific decisions were made, reducing trust and acceptance among authorities.

Fragmented System Architectures

Existing solutions often address individual components such as prediction or communication but lack an integrated, unified framework.

How the Proposed System Overcomes These Limitations:

End-to-End Agentic AI Framework:

The proposed system introduces multiple autonomous agents that handle prediction, assessment, allocation, communication, and learning in a coordinated manner.

Continuous Learning Mechanism:

Reinforcement learning and feedback-based retraining enable the system to improve decisions after every disaster.

Reduced Human Dependency:

The system autonomously performs critical tasks while still allowing human-in-the-loop control for oversight.

Explainable AI Integration:
The system provides decision explanations using confidence scores and reasoning modules, improving transparency and trust.

Unified and Scalable Architecture:
All components are integrated into a single full-stack framework, ensuring seamless data flow and coordination.

METHEDOLOGY:

The methodology followed in this project adopts a **systematic, modular, and iterative approach** to design, develop, and evaluate the Agentic AI-based disaster response system. The complete workflow is divided into six major phases, each contributing to the overall intelligence and adaptability of the system.

Phase 1: Data Acquisition

The first phase involves collecting data required for disaster prediction and emergency assessment. Data is obtained from multiple sources to ensure diversity and reliability. This includes historical disaster datasets, weather data, simulated sensor inputs, emergency reports, and citizen-generated text data. The data consists of both structured data (numerical values such as rainfall, temperature, response times) and unstructured data (textual emergency messages and images). This multi-source data collection enables the system to capture real-world complexity and variability.

Phase 2: Data Preprocessing and Feature Engineering

Raw data collected from various sources often contains noise, missing values, and inconsistencies. In this phase, data preprocessing techniques such as cleaning, normalization, handling missing values, and encoding categorical variables are applied. For textual data, tokenization, stop-word removal, and vectorization techniques are used. Image data is resized, normalized, and enhanced for computer vision models. Feature engineering is performed to extract meaningful attributes that improve model performance and accuracy.

Phase 3: Disaster Prediction Module

In this phase, machine learning models are trained to predict the likelihood and severity of disasters. Supervised learning techniques such as decision trees, random forests, and deep learning models are used based on the nature of the data. The Disaster Prediction Agent continuously analyzes incoming data and generates risk scores for different urban regions. Threshold-based logic is applied to trigger early warnings when risk levels exceed predefined limits. This phase enables proactive disaster preparedness.

Phase 4: Situation Assessment and Severity Analysis

Once a potential disaster or emergency is detected, the Situation Assessment Agent evaluates the real-time conditions. Natural Language Processing techniques are used to analyze emergency messages, reports, and citizen inputs to identify urgency, intent, and emotional tone. Computer Vision models process images or video frames to detect visual indicators such as fire, smoke, flooding, or infrastructure damage. The agent assigns a severity score that determines the priority level of the emergency.

Phase 5: Resource Allocation and Decision-Making

The Resource Allocation Agent is responsible for optimizing the deployment of emergency resources. Reinforcement learning algorithms model the environment as states, actions, and rewards. The agent considers factors such as distance, availability, capacity, severity, and response time to select optimal actions. Rewards are designed to minimize response time and maximize coverage efficiency. Over time, the agent learns improved allocation strategies based on feedback from previous emergencies.

Phase 6: Communication, Execution, and Feedback Learning

In the final phase, the Communication Agent disseminates alerts, instructions, and coordination messages to emergency authorities, hospitals, NGOs, and citizens in real time. A chatbot interface provides guidance and assistance to the public. After the emergency event concludes, the Learning and Feedback Agent analyzes system performance by comparing predicted outcomes with actual results. This feedback is used to retrain models and update agent policies, ensuring continuous improvement and adaptability.

SOFTWARE REQUIREMENTS:

The software requirements define the functional behavior and quality attributes of the proposed Agentic AI–based Disaster Response and Urban Emergency Management System. These requirements ensure that the system operates reliably, securely, and efficiently under real-time emergency conditions.

8.1 Functional Requirements

Disaster Data Collection and Management

The system shall collect and store disaster-related data from multiple sources such as historical datasets, weather APIs, emergency reports, and simulated sensor data. It shall support both structured and unstructured data formats.

Disaster Prediction Module

The system shall analyze incoming data using machine learning models to predict disaster probability, type, and severity. It shall generate risk scores and early warning alerts for high-risk zones.

Situation Assessment and Severity Analysis

The system shall process real-time emergency messages using Natural Language Processing techniques and analyze images using Computer Vision to determine urgency and impact levels.

Resource Allocation and Optimization

The system shall intelligently allocate emergency resources such as ambulances, fire trucks, police units, hospitals, and shelters using reinforcement learning algorithms.

Real-Time Communication and Alerts

The system shall provide real-time alerts and notifications to emergency authorities, hospitals, NGOs, and citizens through dashboards and messaging services.

Citizen Assistance Module

The system shall provide a chatbot interface to assist citizens with emergency guidance, first-aid instructions, shelter locations, and safety measures.

Explainable AI and Decision Transparency

The system shall provide explanations for predictions and allocation decisions to ensure transparency and accountability.

Post-Disaster Analysis and Learning

The system shall analyze response outcomes and retrain models to improve future decision-making.

8.2 Non-Functional Requirements

Scalability

The system shall handle increasing volumes of data, users, and emergency events without performance degradation.

Reliability and Availability

The system shall operate continuously with minimal downtime, especially during critical emergency periods.

Performance and Low Latency

The system shall process data and generate decisions in near real time to ensure timely response.

Security

The system shall enforce authentication, authorization, and secure data storage to protect sensitive information.

Explainability and Trust

The system shall ensure that AI decisions are interpretable to users and authorities.

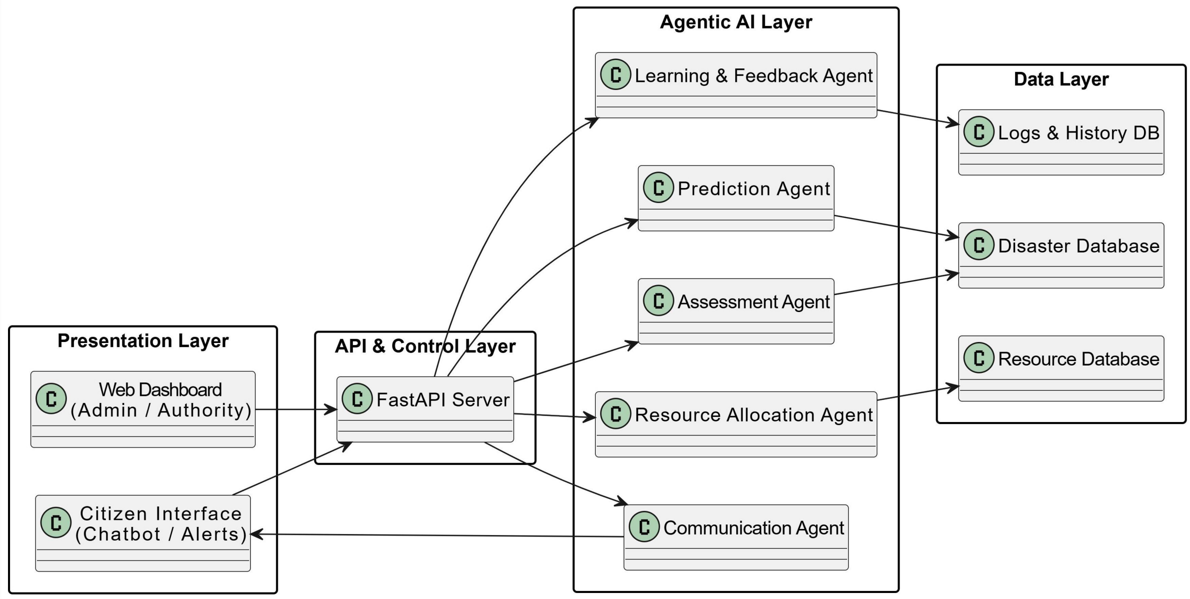
Fault Tolerance

The system shall continue functioning even if one or more modules fail.

Maintainability

The system shall be modular and easy to update, modify, or extend.

SYSTEM ARCHITECTURE:



The proposed system architecture follows a **layered and modular design** to support scalability, maintainability, and real-time performance. The architecture is divided into four primary layers: Presentation Layer, API and Control Layer, Agentic AI Layer, and Data Layer. Each layer has a clearly defined role and communicates with other layers through well-defined interfaces.

The **Presentation Layer** serves as the user interaction component of the system. It includes a web-based dashboard for administrators and emergency authorities, as well as a chatbot interface for citizens. The dashboard displays real-time disaster maps, risk indicators, resource availability, and analytics reports. The chatbot allows citizens to report emergencies, receive alerts, and access safety guidance. This layer communicates with the backend through secure REST APIs and WebSocket connections for real-time updates.

The **API and Control Layer** acts as the central communication hub of the system. It is implemented using a Python-based FastAPI framework, which handles incoming requests from the presentation layer, validates data, and routes requests to the appropriate AI agents. This layer manages authentication, authorization, and role-based access control. It also orchestrates agent interactions by triggering specific agents based on system events such as disaster alerts or emergency reports.

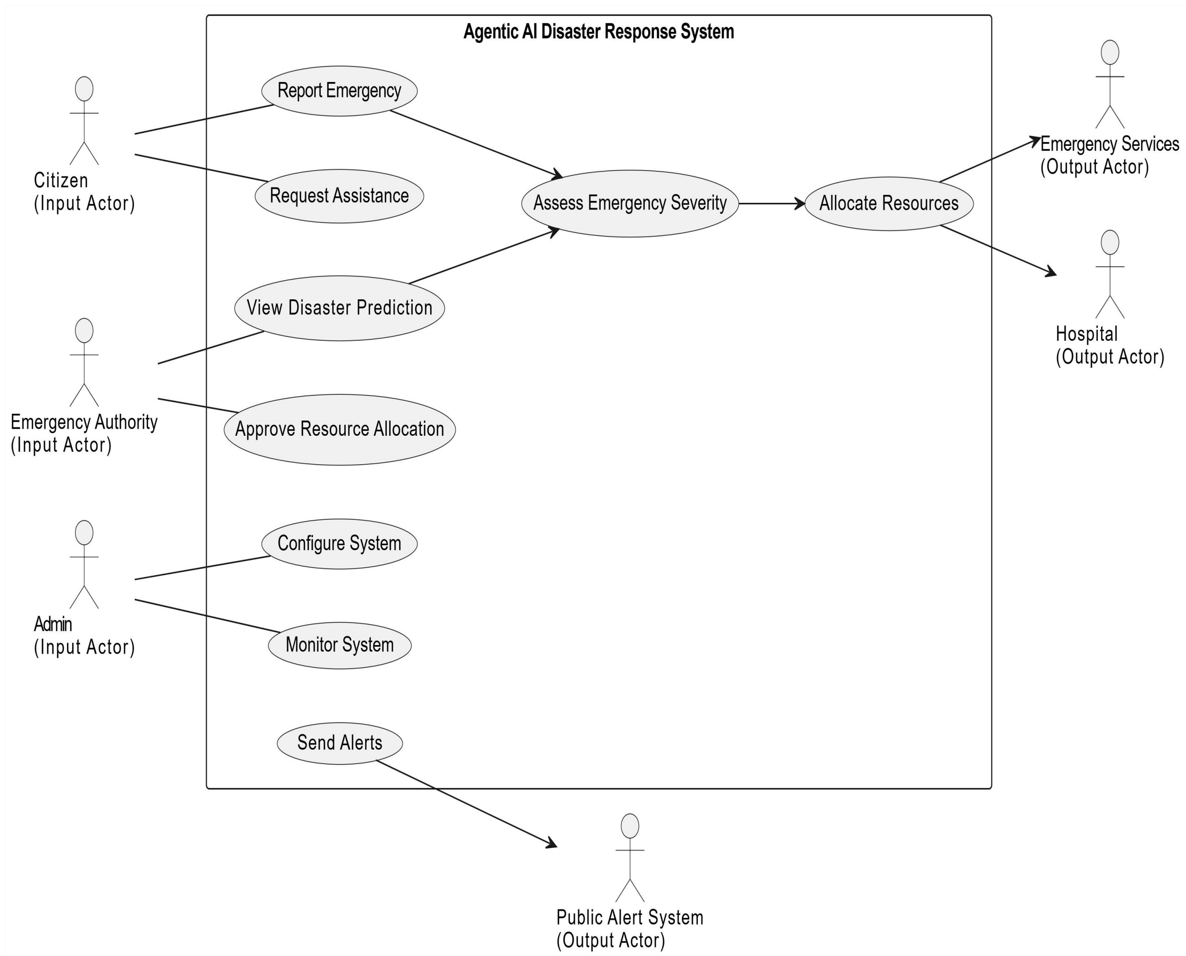
The **Agentic AI Layer** is the core intelligence of the system. It consists of multiple autonomous AI agents, each designed for a specific task. The Disaster Prediction Agent analyzes incoming data to estimate disaster risk. The Situation Assessment Agent evaluates emergency severity using NLP and computer vision techniques. The Resource Allocation Agent optimizes the deployment of emergency resources using reinforcement learning. The Communication Agent handles alert dissemination and coordination, while the Learning and Feedback Agent evaluates system performance and updates models. Agents communicate asynchronously through shared memory or messaging mechanisms, enabling parallel and efficient decision-making.

The **Data Layer** is responsible for data storage and management. Structured data such as user information, disaster records, and resource details are stored in relational databases, while unstructured data such as text logs, images, and agent decisions are stored in NoSQL databases. Caching mechanisms are used to support fast access to frequently used data. This layer ensures data persistence, consistency, and availability across system components.

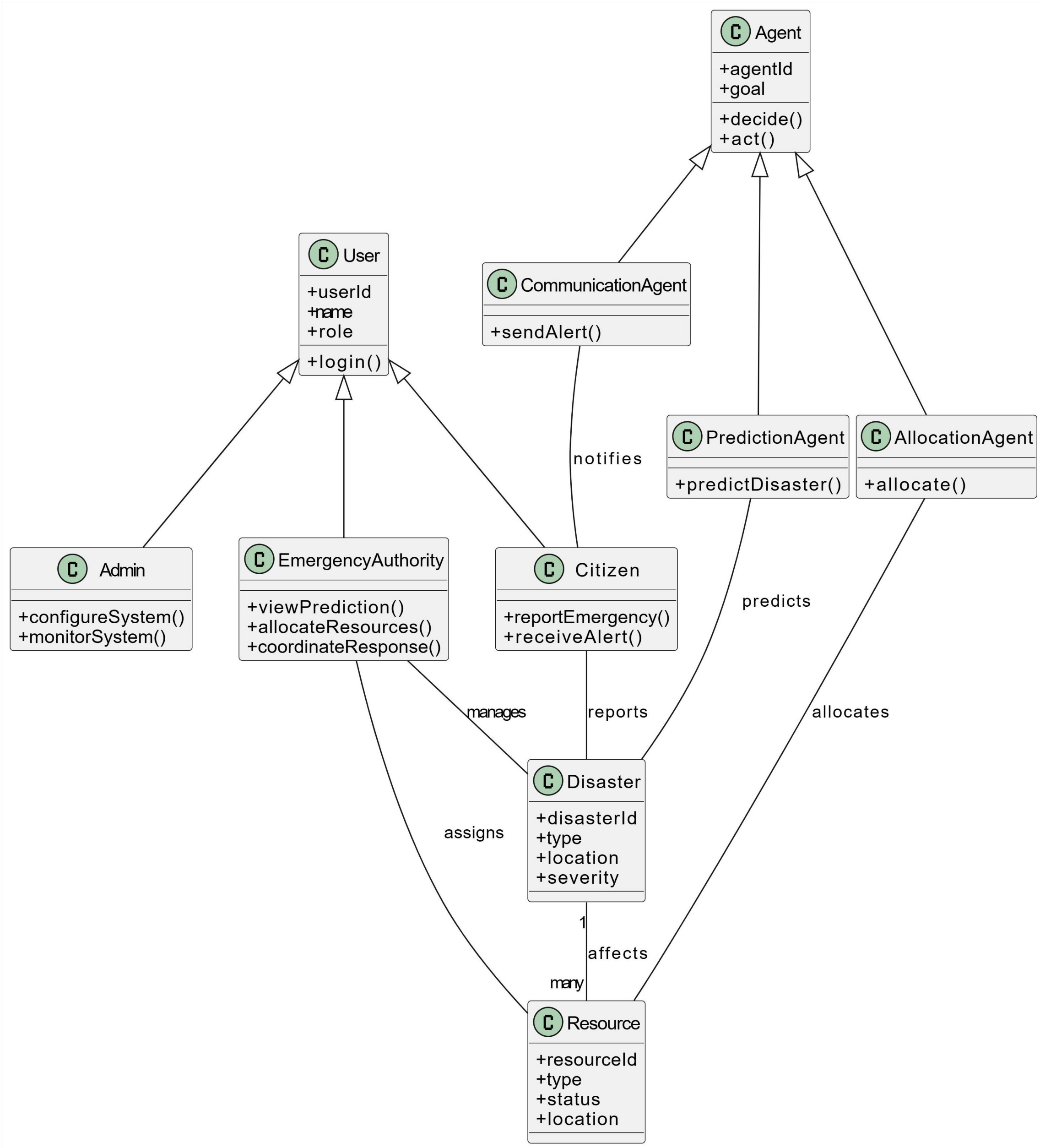
Overall, this layered architecture ensures clear separation of concerns, supports real-time processing, enables autonomous yet explainable decision-making, and allows easy extension for future smart city integrations.

UML DIAGRAMS:

Use Case Diagram



Class Diagram



IMPLEMENTATION

PROJECT PLAN

The AI Disaster Monitoring System is a web-based platform designed to monitor disasters, simulate disaster scenarios using artificial intelligence, and assist authorities in coordinating emergency response actions. The system combines AI-based disaster prediction, geographic visualization, and real-time alerts to provide a centralized disaster monitoring dashboard. The main objective of the project is to improve disaster awareness, response time, and coordination between authorities and the public during emergency situations. By integrating AI agents and real-time monitoring tools, the system helps identify disaster risks, notify users, and assist rescue teams in responding to emergencies efficiently.

The architecture of the system follows a modular full-stack structure consisting of a frontend dashboard, backend AI agents, a database layer, and external datasets used for disaster simulation. The frontend is developed using Streamlit, which provides an interactive web interface for visualizing disaster information. The backend contains multiple AI agents written in Python, each responsible for a specific task in the disaster detection and response process. The system also uses an SQLite database to store alerts, help requests, emergency broadcasts, and user authentication data. Several datasets, including disaster records, hospital locations, and shelter locations, are used to simulate real-world disaster scenarios and support emergency decision making.

The system operates through an AI agent pipeline that simulates disaster events and evaluates their severity. When a simulation is triggered, the system selects disaster information from the dataset and processes it through multiple intelligent agents. The Weather Agent monitors environmental conditions and detects potential risks. The Prediction Agent analyzes disaster parameters and predicts the severity level. The Assessment Agent evaluates the prediction results and determines the final disaster level. The Alert Agent then generates alerts that are displayed on the dashboard. The system stores all disaster alerts and related information in the database so they can be visualized on the monitoring dashboard.

The public dashboard allows users to monitor disasters through a global disaster map that displays disaster locations and alert levels. It also includes real-time alert notifications, emergency broadcast messages, and an AI activity panel that shows which agents are currently analyzing disaster data. In addition to monitoring, the system also allows affected individuals to send help requests through the public portal. Users can submit their location, a short emergency message, and optionally upload an image showing the damage. These requests are stored in the database and appear in the admin command center.

The admin portal acts as a disaster command center where authorities can analyze disasters, run AI simulations, allocate resources, respond to help requests, and send emergency broadcast messages to the public dashboard. Through this centralized control panel, administrators can coordinate rescue operations and communicate important safety information to users. Overall, the system demonstrates how artificial intelligence and real-time data visualization can be combined to build an intelligent disaster monitoring and response platform.

SAMPLE CODE

Alert Agent:

```
import pandas as pd

class AlertAgent:

    def __init__(self):

        # Load alerts dataset
        self.alerts = pd.read_csv("datasets/clean_alerts_dataset.csv")

    def get_random_alert(self):

        alert = self.alerts.sample(1).iloc[0]

        return {
            "disaster_type": str(alert["disaster_type"]),
            "alert_level": str(alert["alert_level"]),
            "country": str(alert["country"]),
```

```
        "severity": float(alert["severity"])
    }
```

Assessment Agent:

```
class AssessmentAgent:

    def assess_disaster(self, weather_risk, prediction, alert_level):

        # Normalize inputs
        weather_risk = weather_risk.lower()
        prediction = prediction.lower()
        alert_level = alert_level.lower()

        # CRITICAL conditions
        if (
            (prediction == "high" and weather_risk == "high")
            or alert_level == "red"
        ):
            return "CRITICAL"

        # HIGH conditions
        if (
            prediction == "high"
            or weather_risk == "high"
        ):
            return "HIGH"

        # MEDIUM conditions
        if (
            prediction == "low" and weather_risk == "medium"
        ):
            return "MEDIUM"
```

```
        # Default
        return "LOW"
```

Communication Agent:

```
class CommunicationAgent:

    def generate_report(self, alert, disaster_level, resources):

        report = {}

        report["disaster_type"] = alert["disaster_type"]
        report["country"] = alert["country"]
        report["alert_level"] = alert["alert_level"]
        report["severity"] = disaster_level

        if "hospital" in resources:

            report["resources"] = {
                "hospital": resources["hospital"],
                "shelter": resources["shelter"],
                "warehouse": resources["warehouse"],
                "transport_route": resources["transport_route"]
            }

        else:

            report["resources"] = "No major deployment required."

        return report
```

Predication agent:

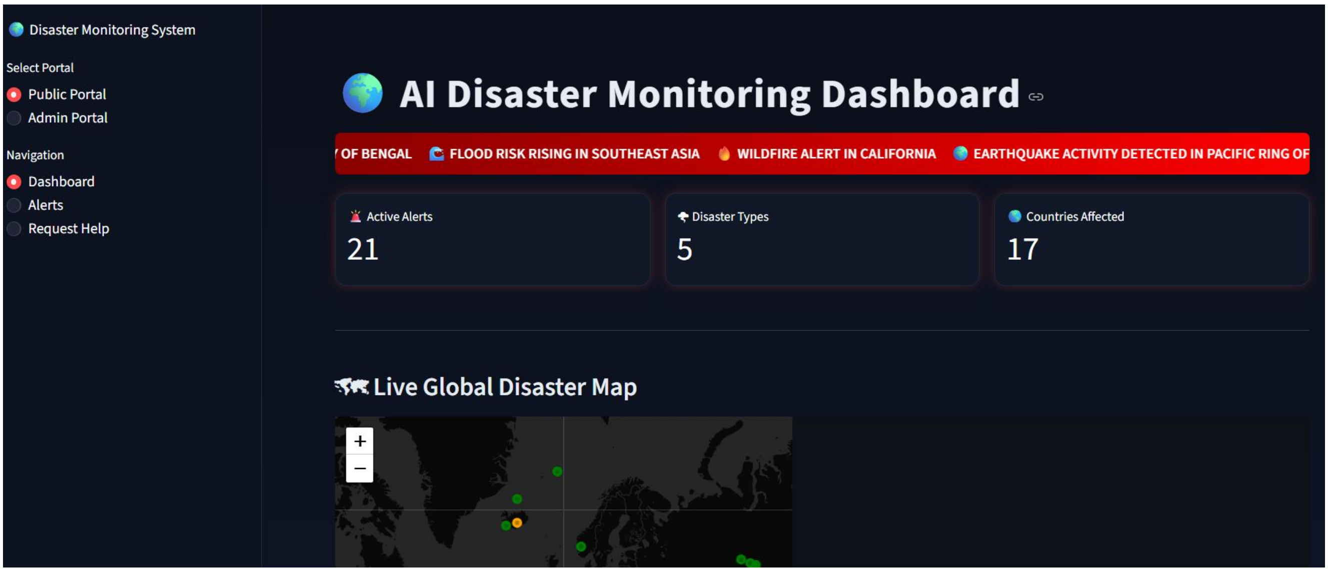
```
import joblib
import pandas as pd


class PredictionAgent:

    def __init__(self):

        # Load trained model
        self.model = joblib.load("models/disaster_prediction_model.pkl")

    # -----------------------------
    # PREDICT DISASTER SEVERITY
    # -----------------------------

    def predict_severity(self, deaths, affected):

        input_data = pd.DataFrame({
            "total_deaths": [deaths],
            "total_affected": [affected]
        })

        prediction = self.model.predict(input_data)

        return prediction[0]
```

Route agent:

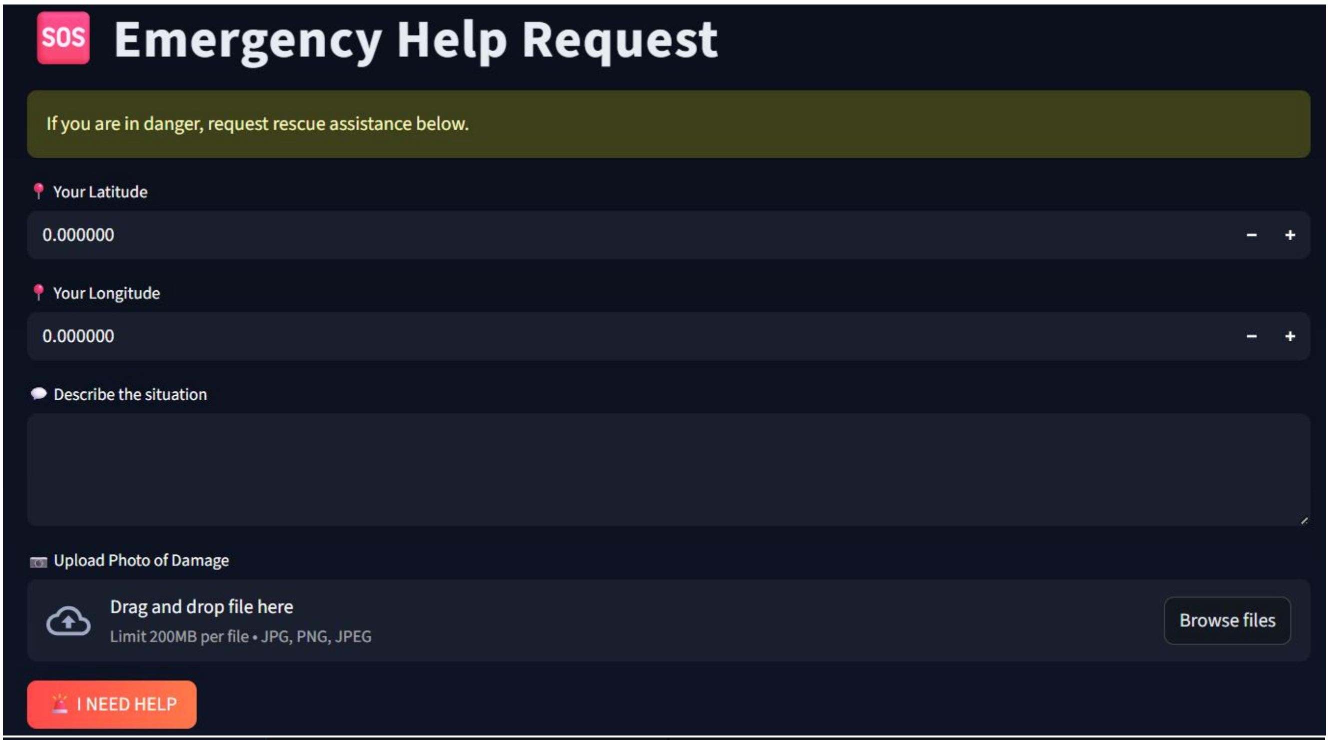
```
import pandas as pd
from math import radians, cos, sin, sqrt, atan2

class RouteAgent:

    def __init__(self):

        self.infrastructure = pd.read_csv(
            "datasets/clean_infrastructure_dataset.csv"
        )

    def calculate_distance(self, lat1, lon1, lat2, lon2):

        R = 6371  # Earth radius

        dlat = radians(lat2 - lat1)
        dlon = radians(lon2 - lon1)

        a = sin(dlat/2)**2 + cos(radians(lat1)) * cos(radians(lat2)) * sin(dlon/2)**2
        c = 2 * atan2(sqrt(a), sqrt(1-a))

        return R * c

    def find_nearest_safe_location(self, disaster_lat, disaster_lon):

        nearest = None
        min_distance = float("inf")

        for _, row in self.infrastructure.iterrows():

            dist = self.calculate_distance(
                disaster_lat,
                disaster_lon,
```

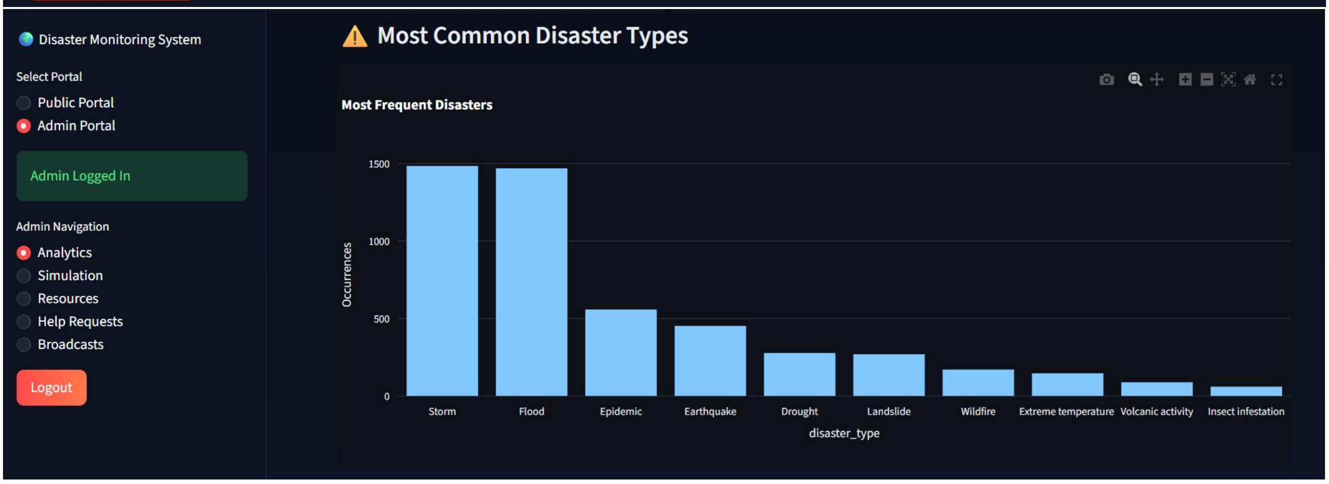
```
            row["latitude"],
            row["longitude"]
        )

        if dist < min_distance:
            min_distance = dist
            nearest = row

    return {
        "name": nearest["name"],
        "type": nearest["infrastructure_type"],
        "latitude": nearest["latitude"],
        "longitude": nearest["longitude"],
        "distance_km": round(min_distance, 2)
    }
```

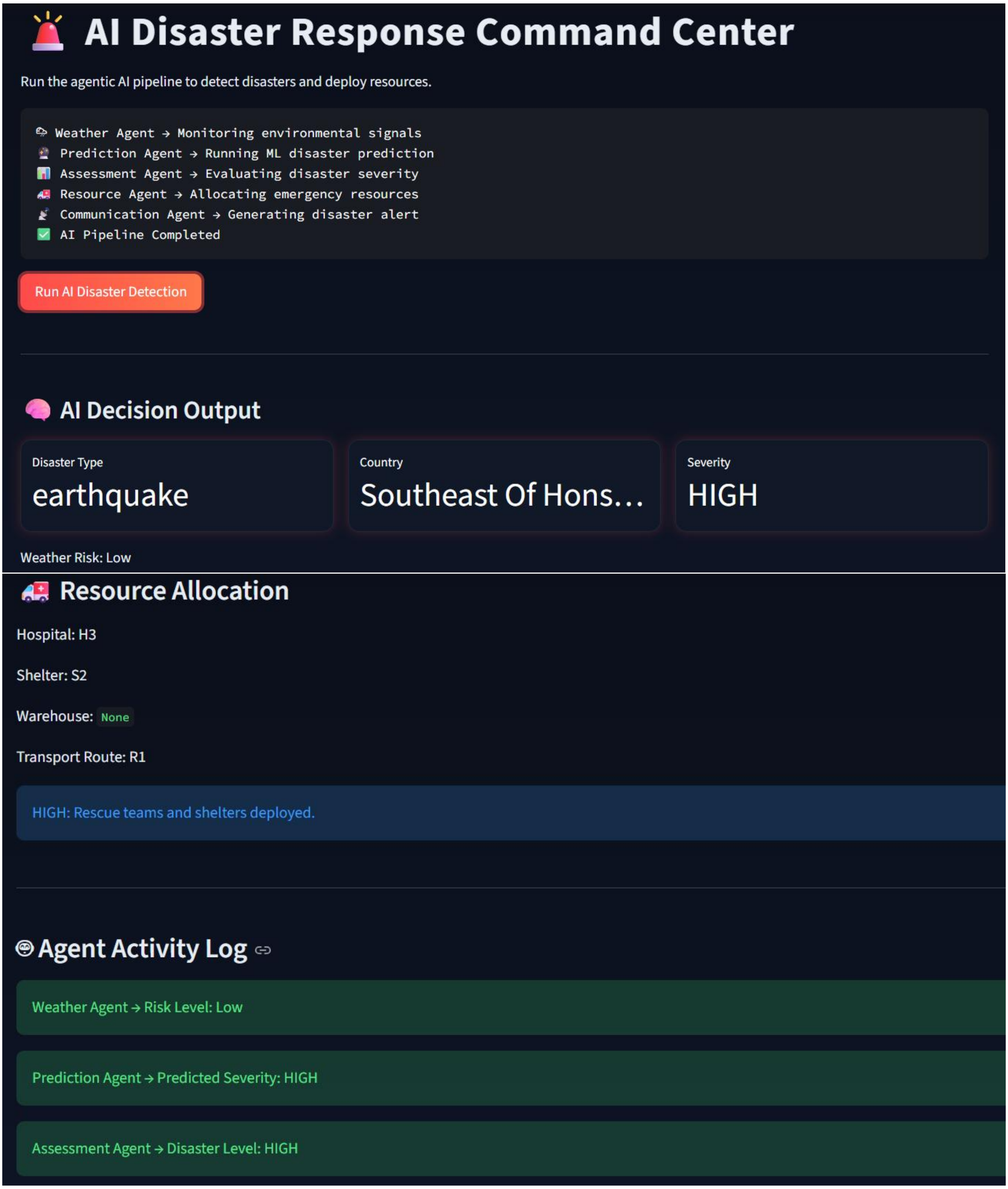
SAMPLE SCREEN SHOTS

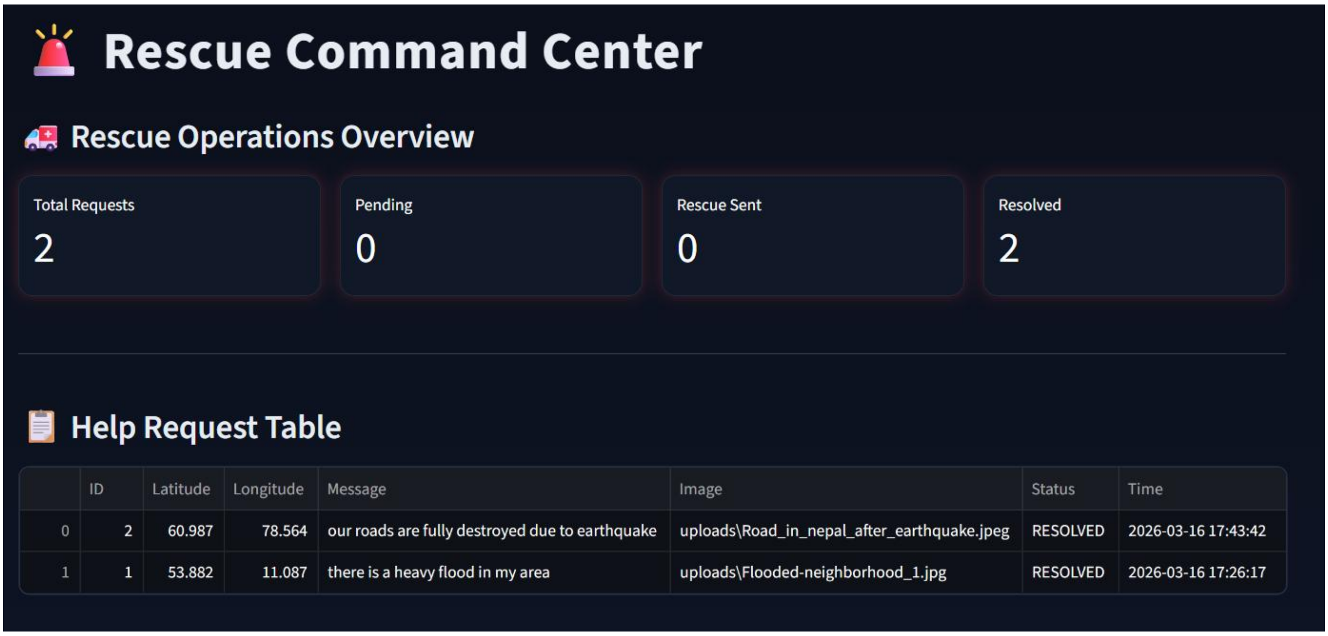



Emergency Broadcasts
VARDA CYCLONE HEADING TO CHENNAI ! | 2026-03-16 18:48:29
Latest Disaster Alerts
forest_fire detected in Russian Federation
eruption detected in Indonesia
flood detected in Spain
eruption detected in Indonesia

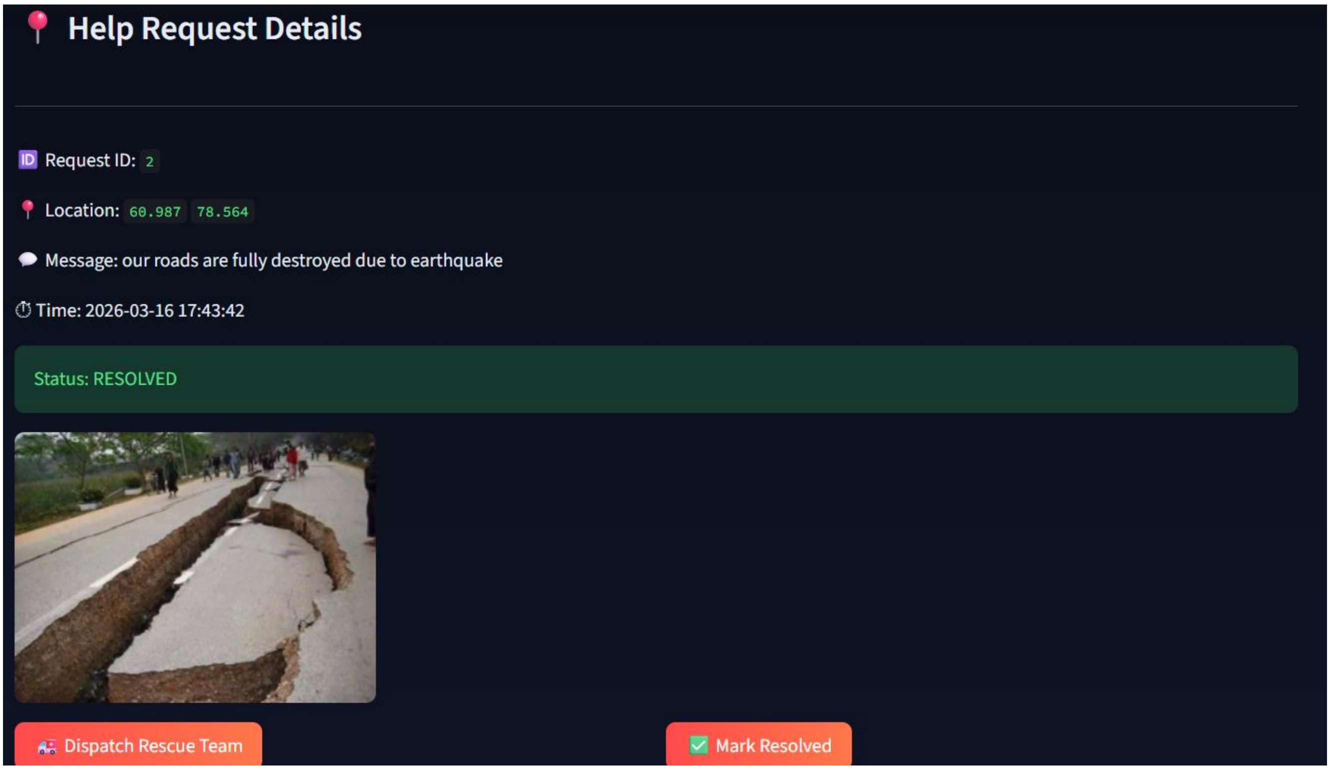

SOS
Emergency Help Request
If you are in danger, request rescue assistance below.
Your Latitude
0.000000
Your Longitude
0.000000
Describe the situation
Upload Photo of Damage
Drag and drop file here
Limit 200MB per file • JPG, PNG, JPEG
Browse files
I NEED HELP

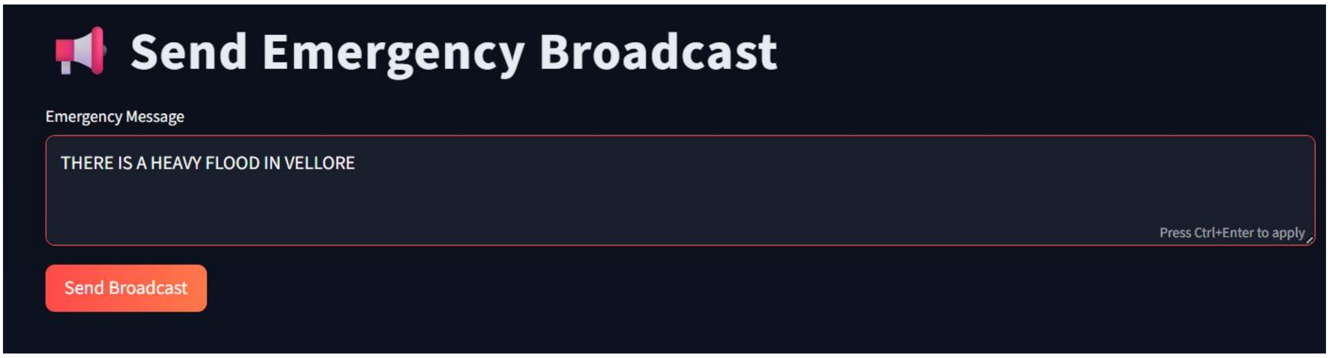

Disaster Monitoring System
Select Portal
Public Portal
Admin Portal
Admin Logged In
Admin Navigation
Analytics
Simulation
Resources
Help Requests
Broadcasts
Logout
Most Common Disaster Types
Most Frequent Disasters
1500
1000
500
0
Occurrences
Storm
Flood
Epidemic
Earthquake
Drought
Landslide
Wildfire
Extreme temperature
Volcanic activity
Insect infestation
disaster_type


AI Disaster Response Command Center
Run the agentic AI pipeline to detect disasters and deploy resources.
Weather Agent → Monitoring environmental signals
Prediction Agent → Running ML disaster prediction
Assessment Agent → Evaluating disaster severity
Resource Agent → Allocating emergency resources
Communication Agent → Generating disaster alert
AI Pipeline Completed
Run AI Disaster Detection
AI Decision Output
Disaster Type
earthquake
Country
Southeast Of Hons...
Severity
HIGH
Weather Risk: Low
Resource Allocation
Hospital: H3
Shelter: S2
Warehouse: None
Transport Route: R1
HIGH: Rescue teams and shelters deployed.
Agent Activity Log
Weather Agent → Risk Level: Low
Prediction Agent → Predicted Severity: HIGH
Assessment Agent → Disaster Level: HIGH


Rescue Command Center
Rescue Operations Overview
Total Requests
2
Pending
0
Rescue Sent
0
Resolved
2
Help Request Table

	ID	Latitude	Longitude	Message	Image	Status	Time
0	2	60.987	78.564	our roads are fully destroyed due to earthquake	uploads\Road_in_nepal_after_earthquake.jpeg	RESOLVED	2026-03-16 17:43:42
1	1	53.882	11.087	there is a heavy flood in my area	uploads\Flooded-neighborhood_1.jpg	RESOLVED	2026-03-16 17:26:17


Help Request Details
Request ID: 2
Location: 60.987 78.564
Message: our roads are fully destroyed due to earthquake
Time: 2026-03-16 17:43:42
Status: RESOLVED
Dispatch Rescue Team
Mark Resolved


Send Emergency Broadcast
Emergency Message
THERE IS A HEAVY FLOOD IN VELLORE
Press Ctrl+Enter to apply
Send Broadcast

SUMMARY:

The rapid growth of urban populations and the increasing frequency of natural and human-induced disasters have made effective disaster response and emergency management a critical challenge for modern cities. Traditional disaster management systems are largely reactive, centralized, and dependent on human decision-making, which often results in delayed responses, inefficient coordination, and poor utilization of emergency resources. This project presented **AGENT-RESCUE**, an Agentic AI–based Disaster Response and Urban Emergency Management System, designed to address these limitations by introducing autonomous, intelligent, and adaptive decision-making capabilities. By leveraging the concept of Agentic AI, the system enables multiple autonomous agents to collaboratively monitor, analyze, and respond to emergency situations in real time.

The proposed system integrates various artificial intelligence techniques including machine learning for disaster prediction, natural language processing for emergency message analysis, computer vision for visual situation assessment, and reinforcement learning for optimal resource allocation. Each agent in the system is assigned a specific responsibility, such as prediction, assessment, allocation, communication, or learning, allowing the system to operate efficiently in complex and dynamic environments. A full-stack Python-based architecture using FastAPI supports real-time communication, secure data handling, and interactive dashboards for authorities and citizens. Explainable AI mechanisms are incorporated to ensure transparency and accountability in decision-making, which is essential for public safety applications. Post-disaster feedback and learning modules allow the system to continuously improve its performance over time.

Overall, AGENT-RESCUE demonstrates how Agentic AI can significantly enhance urban disaster preparedness, response efficiency, and resilience. The system reduces response time, improves coordination among emergency services, and supports informed decision-making during critical situations. Its modular and scalable architecture allows future integration with smart city infrastructure such as IoT sensors, satellite data, drones, and mobile platforms. This project not only contributes a practical solution for urban emergency management but also provides a strong foundation for future research and real-world deployment of autonomous AI systems in disaster response.

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